

ADITYA SHIRSATRAO

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Solapur, Maharashtra, India

OBJECTIVE

Software engineering undergraduate with strong fundamentals in distributed systems, data structures, and object-oriented programming, and hands-on experience building low-latency, event-driven backends at scale. Seeking Goldman Sachs's Software Engineering Summer Internship to contribute to high-performance financial technology systems in a collaborative, risk-focused environment.

EDUCATION

N. K. Orchid College of Engineering & Technology

Solapur, Maharashtra

Bachelor of Technology in Artificial Intelligence and Data Science – CGPA: 8.5/10

Aug. 2023 – Present (Expected 2027)

Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Distributed Systems, OOP, Software Engineering, Machine Learning, Computer Networks.

EXPERIENCE

Fullstack Web Development Intern

Unified Mentor Pvt. Ltd

Distributed Systems & Backend Engineering

Jul. 2025 – Jan. 2026

- Designed a **cloud-native microservices e-commerce platform** with payment orchestration, inventory consistency, and distributed caching using an event-driven architecture on Docker and Kubernetes.
- Built an **AI-powered code review platform** (Kafka + Redis, **45%** lower latency via parallel workers) and an **event streaming analytics system** (Kafka + Spark Streaming) for real-time dashboards and alerting.
- Engineered a **real-time collaborative IDE** with sub-100ms synchronization using WebSockets, Socket.IO, Redis pub/sub, and CRDT-based operational transforms.

AI/ML Lead

Google Developer Groups on Campus (GDG-NKOCET)

Leadership & Mentorship

Sep. 2025 – Present

- Lead technical workshops and hackathon mentorship for a **200+ student** community, emphasizing clean architecture, testing, and system design.

PROJECTS

Order Matching Engine | C++17, Price-Time Priority, GTC/IOC/FOK, FastAPI, WebSockets, Redis | Code Jun. 2026 – Jul. 2026

- Engineered a **dependency-free C++17 limit order book** with price-time priority and FIFO per level, supporting LIMIT/MARKET/STOP/STOP-LIMIT and GTC/IOC/FOK semantics using deterministic integer quantities (no floats).
- Benchmarked matching at **10.9M orders/sec** (0.09µs/order); exposed trade tape + depth over a FastAPI/WebSocket feed; 9

Tracefly – Distributed Observability Platform | FastAPI, React, Redis, PostgreSQL, Multi-language SDKs | Code Apr. 2026

- Built a multi-tenant error-tracking platform processing high-volume events with Redis caching and async workers; multi-language SDK (Python/Java/Kotlin/Go/C++/JS).

SentinelX – Real-Time Risk Scoring Platform | FastAPI, Django, Celery, Redis, PostgreSQL | Code May 2026

- Designed RBAC, CORS, and a health-checked deployment with GitHub Actions CI + PyPI publishing.
- Shipped a weighted multi-signal risk engine with async task queues, JWT auth, and RBAC — production-shaped backend demonstrating correctness, reliability, and auditability.
- Win: **1st Place** Hack to Future 3.0 (600+ teams).

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript (ES6+), TypeScript, SQL

Backend & Distributed Systems: FastAPI, Node.js, Express.js, Redis, Kafka, Message Queues, Microservices, REST APIs, WebSockets

Infrastructure: Docker, Kubernetes, GitHub Actions CI/CD, AWS (EC2/S3/Lambda), Linux, Grafana/Prometheus

Core Concepts: DSA (366+ LeetCode), System Design, OOP, Concurrency & Asynchronous Programming, Databases, Security (RBAC, JWT)

ACHIEVEMENTS & CERTIFICATIONS

Competitive Programmer: 366+ LeetCode problems (104 Easy, 176 Medium, 86 Hard) + 50 SQL problems.

IEEE First-Author Publication at ICCTWC 2026; **1st Place** Hack to Future 3.0 (600+ teams); **1st Runner-Up** Orchathon 2K26 (800+ teams).

Open Source: 50+ PRs merged across Google (OSS-Fuzz), Microsoft, Apache (DataFusion), and Hedera (Hiero SDK).

Certifications: JPMorgan Chase & Electronic Arts SWE Virtual Experience; NPTEL Java (Elite+Gold, 91/100); Kaggle AI Agents.

Leadership: Class Representative (120+ cohort); seminar speaker on AI and Chandrayaan-3.